



Vibeathon 6.0 (Vibecoding Hackathon) - July

Brigade — the menu as a live function of the pantry



Team Name: bishalranabr7

Team Leader: Bishal Rana

College: Parala Maharaja Engineering College

Year & Department: CSE — 2023 batch

Problem Statement: Smart Restaurant Management System

Live now: <https://brigade-flame.vercel.app>

Code: <https://github.com/bishalbar77/brigade>

Sign in as owner@brigade.test / grill@brigade.test / priya@brigade.test

Password for every demo account: brigade-demo-2026





CURRENT PROBLEM

A menu promises what stock cannot keep



Diners, cooks and owners all guess



Asked verbally, stale when answered



Tills already auto-86 — none predict



4-10% food wasted; cost 28-32%



Proposed Solution



- Runway: minutes until a dish 86s
- portions ÷ tonight's real sell rate
- Menu, ordering, KDS, runway board
- Availability reaches the diner
- Predicted, not reported — time to act



TECHNICAL APPROACH



- Next.js 15, TypeScript, Supabase
- Availability computed, never stored
- Append-only ledger; 43 RLS policies
- No LLM – deterministic statistics
- 70 tests · 27 DB · 120 end-to-end





Use Cases & Impact

- A QR scan shows what is available
- Seven roles, enforced in Postgres
- Independent restaurants; multi-tenant
- Less waste, honest waiting times
- Every figure comes from the ledger



Future Scope & Conclusion

- Surface notifications, recommendations
- Supplier APIs, payments, accounting
- Multi-tenant — a second site is a row
- Runway suits any perishable stock
- Tills say a dish IS out. We say when.